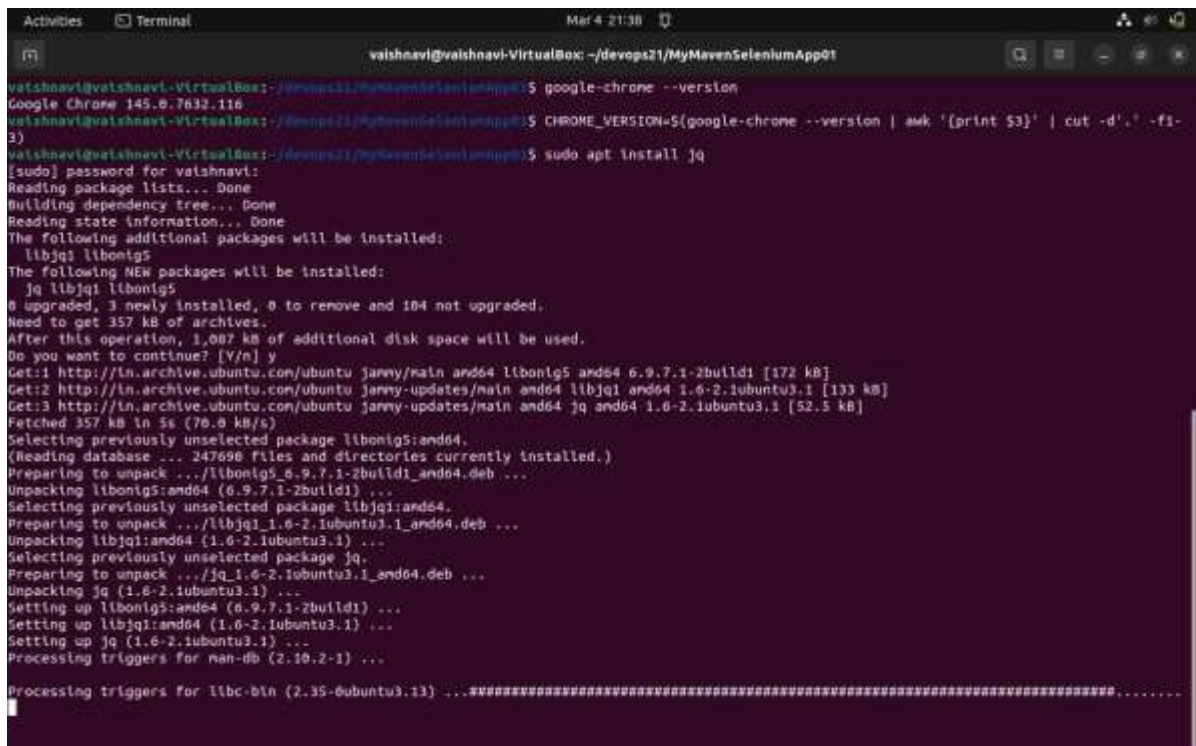


Maven Project To Connect To Web Servers Using Selenium Dependency

Step1:

Before beginning the project building, install ChromeDriver.

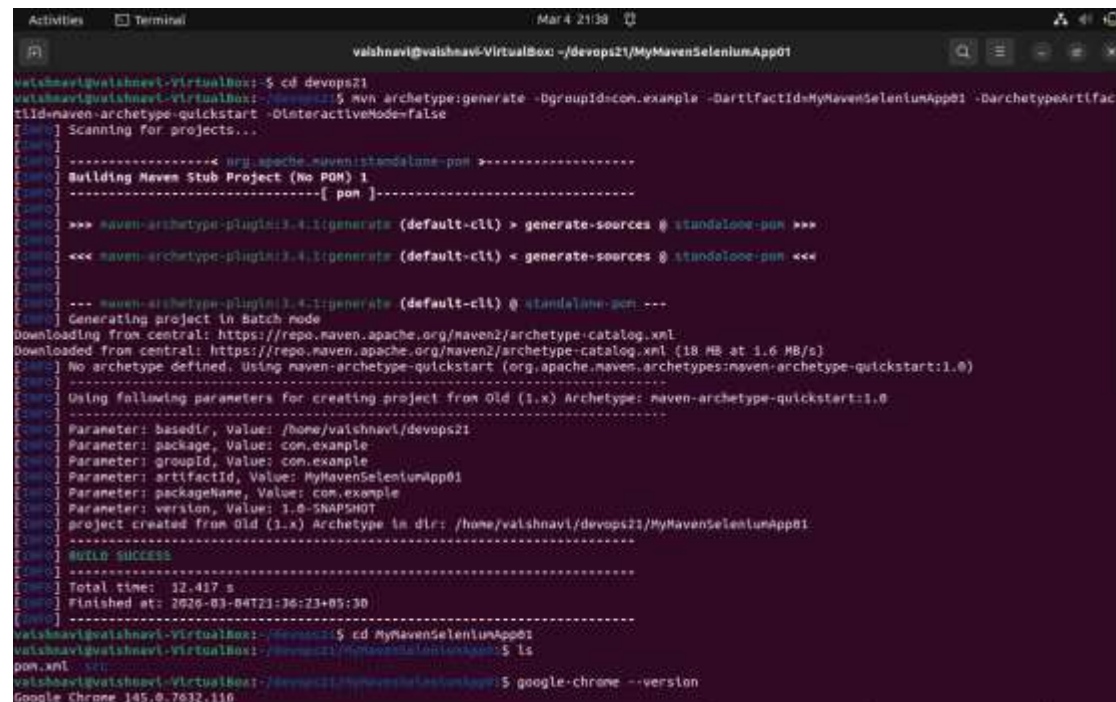
- Check the Google Chrome version: `google-chrome --version` Example output: Google Chrome 133.0.6943.141
- Extract the Chrome version and store it in a variable: `CHROME_VERSION=$(google-chrome --version | awk '{print $3}' | cut -d'.' -f1-3)`
- Install jq (used to process JSON data): `sudo apt install jq`
- If curl is not installed, install it using: `sudo apt install curl`
- Get the latest compatible ChromeDriver version: `LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latestpatch-versions-per-build.json" | jq -r ".builds["$CHROME_VERSION"].version")`
- Download ChromeDriver: `wget https://storage.googleapis.com/chrome-for-testing/public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip`
- Extract the downloaded file: `unzip chromedriver-linux64.zip`
- Move ChromeDriver to the system path: `sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver`
- Give execution permission: `sudo chmod +x /usr/local/bin/chromedriver`
- Verify the installation: `chromedriver --version`
- The output version should match the installed Chrome version



```
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp01
vaishnavi@vaishnavi-VirtualBox:~/devops21/MyMavenSeleniumApp01$ google-chrome --version
Google Chrome 145.0.7632.116
vaishnavi@vaishnavi-VirtualBox:~/devops21/MyMavenSeleniumApp01$ CHROME_VERSION=$(google-chrome --version | awk '{print $3}' | cut -d'.' -f1-3)
vaishnavi@vaishnavi-VirtualBox:~/devops21/MyMavenSeleniumApp01$ sudo apt install jq
[sudo] password for vaishnavi:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libjq1 libonig5
The following NEW packages will be installed:
  jq libjq1 libonig5
0 upgraded, 3 newly installed, 0 to remove and 104 not upgraded.
Need to get 357 kB of archives.
After this operation, 1,007 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ln.archive.ubuntu.com/ubuntu jammy/main amd64 libonig5 amd64 6.9.7.1-2build1 [172 kB]
Get:2 http://ln.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libjq1 amd64 1.6-2.1ubuntu3.1 [133 kB]
Get:3 http://ln.archive.ubuntu.com/ubuntu jammy-updates/main amd64 jq amd64 1.6-2.1ubuntu3.1 [52.5 kB]
Fetched 357 kB in 5s (70.0 kB/s)
Selecting previously unselected package libonig5:amd64.
(Reading database ... 247698 files and directories currently installed.)
Preparing to unpack .../libonig5_6.9.7.1-2build1_amd64.deb ...
Unpacking libonig5:amd64 (6.9.7.1-2build1) ...
Selecting previously unselected package libjq1:amd64.
Preparing to unpack .../libjq1_1.6-2.1ubuntu3.1_amd64.deb ...
Unpacking libjq1:amd64 (1.6-2.1ubuntu3.1) ...
Selecting previously unselected package jq.
Preparing to unpack .../jq_1.6-2.1ubuntu3.1_amd64.deb ...
Unpacking jq (1.6-2.1ubuntu3.1) ...
Setting up libonig5:amd64 (6.9.7.1-2build1) ...
Setting up libjq1:amd64 (1.6-2.1ubuntu3.1) ...
Setting up jq (1.6-2.1ubuntu3.1) ...
Processing triggers for man-db (2.10.2-1) ...
```

Step 2:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```



```

Activities Terminal Mar 4 21:38
valshnavi@valshnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp01

valshnavi@valshnavi-VirtualBox:~$ cd devops21
valshnavi@valshnavi-VirtualBox:~/devops21$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> mvn-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< mvn-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 KB at 1.6 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/valshnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Old (1.x) Archetype in dir: /home/valshnavi/devops21/MyMavenSeleniumApp01
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO]
[INFO] Total time: 32.417 s
[INFO] Finished at: 2026-03-04T21:36:23+05:30
[INFO]
[INFO] -----
valshnavi@valshnavi-VirtualBox:~/devops21$ cd MyMavenSeleniumApp01
valshnavi@valshnavi-VirtualBox:~/devops21/MyMavenSeleniumApp01$ ls
pom.xml
valshnavi@valshnavi-VirtualBox:~/devops21/MyMavenSeleniumApp01$ google-chrome --version
Google Chrome 145.0.7632.110

```

The command can be broken down as:

`DgroupId=com.example` – project’s base package (reverse-domain style).

`-DartifactId=MyMavenSeleniumApp` – defines the name of the project (will also be the folder name).

`-DarchetypeArtifactId=maven-archetype-quickstart` – sets the Maven template to be the basic Java application.

`-DinteractiveMode=false` – deactivates interactive mode and uses the given values automatically.

Step 2:

Use `$cd` command to go into the directory.

`$gedit pom.xml`

`pom.xml` file is created to configure the Maven project. It defines the project information, adds required dependencies like Guava and Commons IO libraries, and includes build plugins to compile the Java code, run unit tests, and package the application into an executable JAR file.

<dependencies>

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>3.8.1</version>
  <scope>test</scope>
</dependency>

<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
  <groupId>org.seleniumhq.selenium</groupId>
  <artifactId>selenium-java</artifactId>
  <version>4.29.0</version>
</dependency>
</dependencies>

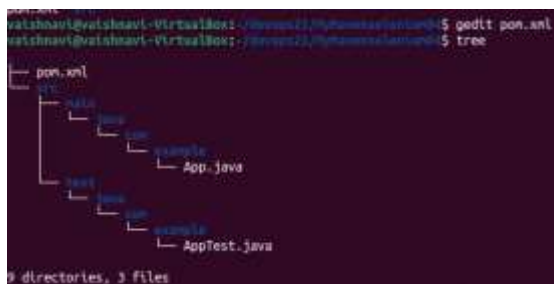
<!-- Build: Configuring plugins and build settings -->
<build>
  <plugins>
    <!-- Example: Maven Compiler Plugin to compile Java code -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.8.1</version>
      <configuration>
        <source>1.7</source>
        <target>1.7</target>
      </configuration>
    </plugin>
    <!-- Example: Maven Surefire Plugin to run tests -->
    <plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>
    <version>3.0.1</version>
    <configuration>
        <archive>
            <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
            </manifestEntries>
        </archive>
    </configuration>
</plugin>

</plugins>
</build>
</project>
```

Execute \$tree command to view the structure of the project



```
vaishnav@vaishnav-VirtualBox: /home/vaishnav/Projects/automation$ cd pom.xml
vaishnav@vaishnav-VirtualBox: /home/vaishnav/Projects/automation$ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   └── com
│   │   │       └── example
│   │   │           └── App.java
│   └── test
│       ├── java
│       │   ├── com
│       │   │   └── example
│       │   │       └── AppTest.java
└── 9 directories, 3 files
```

Step 3:

inspect the SauceDemo webpage to find the element IDs required for automation.

Right-click on the webpage and select Inspect. The HTML code will be displayed. Click on the cursor icon (arrow symbol) available on the top-left of the inspect panel, then hover over the username field to identify its ID, which is user-name.

Similarly, check the password field (ID: password) and the login button (ID: login-button). Make a note of these IDs, as they will be used in the App.java file.

Next,

Edit the App.java file

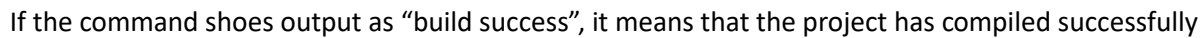
```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

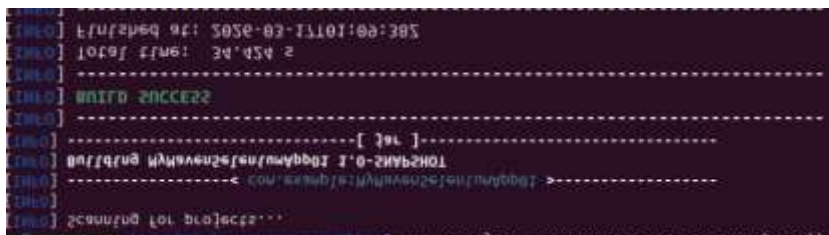
/** * Hello world! * */

public class App
{
    public static void main( String[] args )
    {
        WebDriver driver=new ChromeDriver();
        driver.get("https://www.saucedemo.com/");
        driver.manage().window().maximize();
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();
    }
}
```

Execute the command `$mvn clean install`



```
$mvn exec:java -Dexec.mainClass="com.example.App"
```

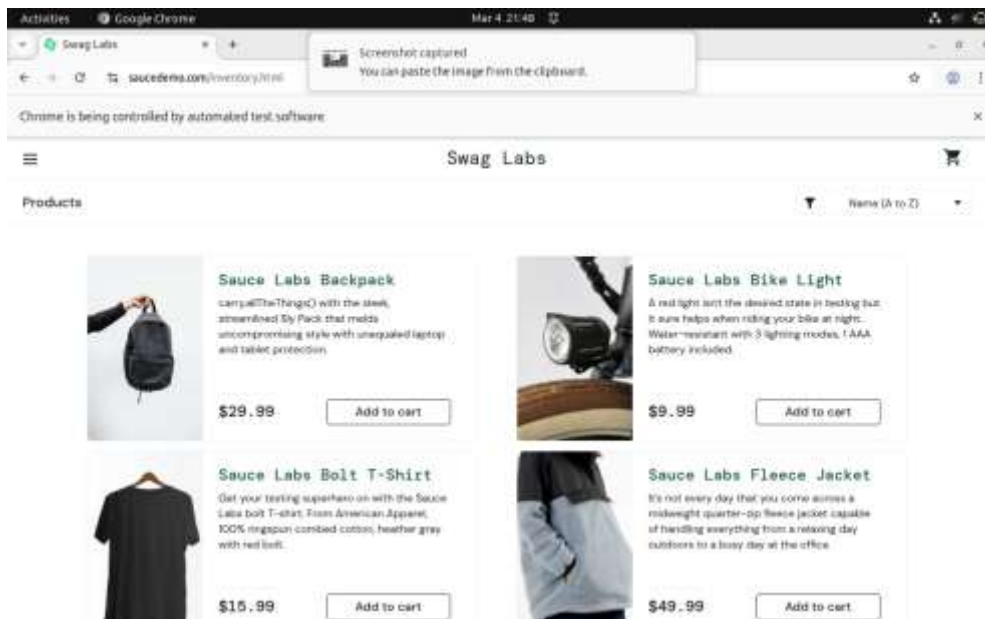
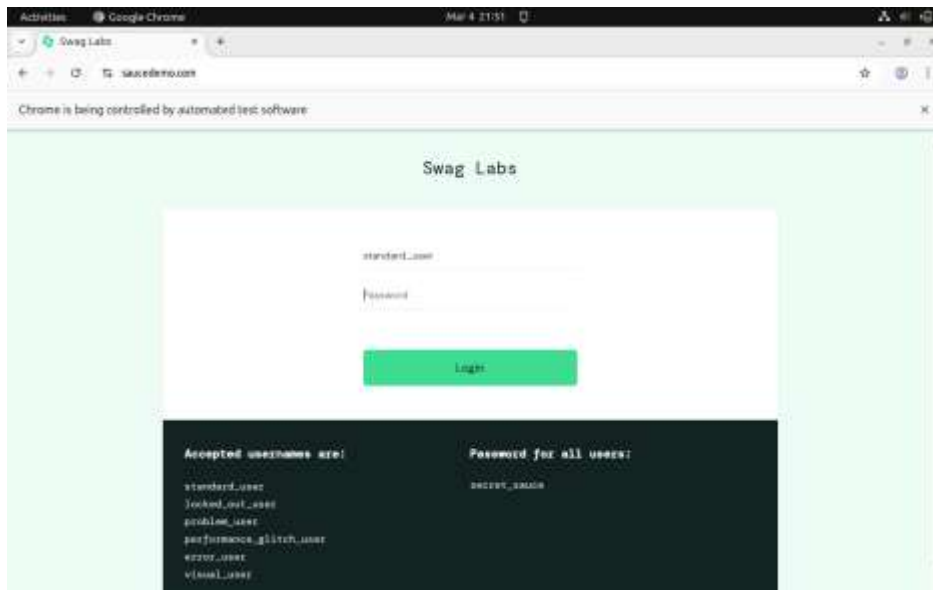


Sauce Demo Website

Sauce Demo (<https://www.saucedemo.com/>) is a fake e-commerce website created by Sauce Labs for testing and learning web automation tools like Selenium.

It is used for:

- Writing automation scripts and UI tests
- Demonstrating cloud testing features
- Learning and practicing test automation



Pushing The Project into Git:

Execute the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$ git init
```

```
$ git add .
```

```
$ git commit -m "initial commit"
```

```
$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp.git
```

```
$ git remote set-url origin https://PAT*@github.com/vaishnavim1121/MyMavenSeleniumApp
```

```
$ git push -u origin master
```

```

$ git branch -m master
Initialized empty Git repository in /home/vaishnavi/devops21/MyMavenSeleniumApp03/.git/
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git add .
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git commit -m "First commit"
[master (root-commit) 1af4a6b] first commit
13 files changed, 265 insertions(+)
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/App.java
create mode 100644 src/test/java/com/example/AppTest.java
create mode 100644 target/MyMavenSeleniumApp03-1.0-SNAPSHOT.jar
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git config --global user.name "vaishnavim1121"
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git config --global user.email "vaishnavim1121@gmail.com"
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp03.git
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ set-url origin https://ghp_NWMyMp5Sxgt2QxFOV7noCjtJaL6ZV3wVIFI@github.com/vaishnavim1121/MySeleniumApp03
set-url: command not found
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git remote set-url origin https://ghp_NWMyMp5Sxgt2QxFOV7noCjtJaL6ZV3wVIFI@github.com/vaishnavim1121/MySeleniumApp03
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git push -u origin master
remote: Repository not found.
fatal: repository 'https://github.com/vaishnavim1121/MySeleniumApp03/' not found
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git remote set-url origin https://ghp_NWMyMp5Sxgt2QxFOV7noCjtJaL6ZV3wVIFI@github.com/vaishnavim1121/MyMavenSeleniumApp03
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ git push -u origin master
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 10 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (39/39), 8.46 KiB | 721.00 KiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/vaishnavim1121/MyMavenSeleniumApp03
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
vaishnavi@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $

```